File 64: Exurban Data Center Hydrology: Water-Use Modeling & Ecological Offsets

> ***Corporate ESG & Hydrologic Siting Standard**: This document establishes the strategic scientific prospectus analyzing exurban data center water consumption and outlining our watershed-scale thermal mitigation offset framework. By modeling evaporative cooling tower water drawdowns and their direct thermal impacts on North Georgia's coldwater spawning headwaters, we build a high-status corporate relationship bridge. This manual details data center hydrologic equations, groundwater thermal stresses, and a structured strategic offset narrative to position Hunter Morris's Spawning Bio-Reserves as the premium ESG investment vehicle for hyperscale data center operators (QTS, Microsoft, Google).*

I. Data Center Evaporative Cooling Water-Use Modeling

As hyperscale data centers expand rapidly across Georgia, their massive water consumption has emerged as a major environmental challenge. While many facilities utilize direct-expansion air-cooling systems, high-density computing clusters require liquid-cooled heat exchangers and **Evaporative Wet Cooling Towers** to maintain stable processor temperatures.

1. The Cooling Tower Mass Balance Equation

We model a data center's total hydrologic water intake (**M_makeup**, or Make-Up Water) using the mass balance equation:

> ***M_makeup = M_evaporation + M_drift + M_blowdown***

- *Where*:
- **M_evaporation** is the pure water vapor evaporated to remove heat (m^3/hr).
- **M_drift** is the water lost as liquid droplets carried away by wind exhaust (typically limited to less than 0.005% of total recirculating water using high-efficiency drift eliminators).
- **M_blowdown** is the wastewater discharged from the basin to prevent mineral scaling as dissolved solids concentrate.

2. Calculating Evaporative Water Loss (M_evaporation)

The water evaporated is directly proportional to the total thermal load (**Q_thermal**) rejected by the servers:

> ***M_evaporation = Q_thermal / (rho_w * lambda_v)***

- *Where*:
- **Q_thermal** is the heat load (typically **90%** of the data center's total IT power capacity, e.g., a **100MW** facility rejects approximately 90 MW of heat).
- **rho_w** is the density of water (1000 kg/m^3).
- **lambda_v** is the latent heat of vaporization of water (approximately 2400 kJ/kg at 25°C).

- **100MW Hyperscale Evaporation Calculation**:
- Heat Load: Q_thermal = 90 MW = 90,000 kW = 90,000 kJ/s
- Evaporation: M_evaporation = 90,000 kJ/s / (1000 kg/m^3 * 2400 kJ/kg) = 0.0375 m^3/s = 135 m^3/hr
- Daily Loss: **855,000 Gallons Per Day**

3. Calculating Blowdown Wastewater (M_blowdown)

To prevent mineral scale deposits from building up on heat exchangers, a portion of the concentrated basin water must be discharged (blown down). This is dictated by the **Cycles of Concentration (CoC)**, representing how many times dissolved solids concentrate:

> ***M_blowdown = M_evaporation / (CoC - 1)***

- *Design Siting Standard*: At an average exurban Georgia CoC of **4.0**, the blowdown water is:
- M_blowdown = 855,000 GPD / (4 - 1) = **285,000 GPD**
- **Total Make-Up Water Requirement**:
- M_makeup = 855,000 GPD + 285,000 GPD = **1,140,000 Gallons Per Day (per 100MW IT load)**
- *(Large multi-building campuses often exceed 300MW, drawing upwards of **3,420,000 Gallons Per Day** directly from exurban aquifers).*

II. Groundwater Table Drawdowns & Headwater Thermal Stresses

When exurban data centers draw millions of gallons of water per day from municipal wells, they create a significant **cone of depression** in the local groundwater aquifer. This drawdown has severe, cascading ecological impacts on headwater trout streams:

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                          [GROUNDWATER DRAWDOWN & THERMAL STRESS IMPACT]

           Data Center Aquifer Wells (3.4M Gallons/Day)
                     ||
                     ||
                     vv (Aquifer Drawdown)
           =====================\\                         //===================
                                 \\   Cone of Depression  //  <=== Lowers Water Table
                                   \\                    //
           ~~~~~~~~~~~~~~~~~~~~~~~~~~\\~~~~~~~~~~~~~~~~~//~~~~~~~~~~~~~~~~~~~~~~~~
           Shallow Stream Bed ( Anderson Creek Spawning Reach)
                                       |
                                       v [Decreased Cold Groundwater Seeps]

           * Cold groundwater flow (54°F) is cut off.
           * Stream water temperature rises above 68°F (Lethal thermal stress).
           * Spawning gravels dry up or fill with algae, suffocating wild trout eggs.
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1. Aquifer Drawdown Physics

According to the **Theis Equation**, the drawdown (**s**) in a confined aquifer surrounding a high-capacity well is modeled as:

> ***s(r, t) = [Q / (4 pi T)] * W(u)***

- *Where*: **Q** is well discharge rate, **T** is aquifer transmissivity, **r** is radial distance, and **W(u)** is the exponential integral (Well Function).
- **Cone of Depression**: As water tables fall within this radius, the natural gradient directing cold groundwater to nearby headwater streams is **reversed or eliminated**.

2. Downstream Thermal Pollution Mechanics

- **Loss of the 54°F (12.2°C) Baseflow**: Headwater trout streams maintain stable temperatures during hot summer months because they are fed by constant 54°F groundwater springs.
- **Thermal Spikes**: When data center well draws lower water tables and cut off these cold springs, stream flows drop and solar radiation heats the remaining shallow water. Stream temperatures spike rapidly above **68°F (20°C)**, exceeding critical wild trout stress thresholds.
- **Egg Mortality**: Spawning gravels experience complete oxygen depletion as water flow slows, suffocating the incubating brook trout eggs and causing local spawning ranges to collapse.

III. The ESG Siting Conflict: Coldwater Conservation vs. Big Tech

As tech giants (Microsoft, Google, Meta) and colocation operators (QTS Data Centers) face intense regulatory scrutiny, their hydrologic footprints have become a primary target for environmental groups and local communities:

1. The Trout Unlimited "Coldwater Conservation" Narrative

Trout Unlimited is nationally recognized for protecting coldwater fisheries. The organization is highly vocal about industrial activities that cause stream temperature increases or aquifer drawdowns.

- **The Siting Risk**: If a data center developer sites a new campus in an exurban watershed that borders native Brook Trout headwaters (such as the Upper Toccoa or Etowah basins), local conservation groups can mount significant public opposition.
- **The Reputational Risk**: A news headline stating *"Data Center Water Use Threatens Georgia's Last Native Spawning Riffles"* can severely impact a corporation's public ESG rating and delay multi-million-dollar zoning approvals for years.

IV. The Blue Ridge Solution: High-Value Spawning Sanctuary Offsets

To resolve this conflict and turn corporate risk into a highly positive conservation narrative, Blue Ridge Stream Restoration has structured a premium B2B **Water and Thermal Offset Framework**.
Hunter Morris and Hadi Irvani present this strategic offset narrative directly to data center sustainability executives:

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                      [DATA CENTER WATERSHED OFFSET VALUE CHAIN]

   Hyperscale Operator       Blue Ridge Spawning Sanctuary      Trout Unlimited & Warnell
  +-----------------------+  +----------------------------------+  +-------------------------
+
  | - Aquifer drawdown   |  | - Regan failing pasture banks  |  | - Joint science research
|
  |   thermal risk       |=>| - 96% canopy solar attenuation |=>| - Positive ESG ratings
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 |
  | - ESG reputational   |  | - Intercept cold seeps        |  | - Fast-track regional
|
  |   vulnerability      |  | - Create deep plunge pools     |  |   zoning approvals
|
  +----------------------+  +-------------------------------+  +------------------------
+
```

The Sourcing Pitch and Strategic Alignment:

1. **The Thermal Offset Credit**: While a data center may consume groundwater in one part of a HUC-8 basin, they can *fully offset* this regional thermal impact by funding a **Blue Ridge Spawning Sanctuary** along a degraded reach in the same watershed.
2. **Turnkey Scientific Verification**: We install our off-grid **LoRa/Meshtastic water quality monitoring stations** on the funded reach. The live telemetry (water temperature, DO, stage) is streamed directly to Trout Unlimited and the UGA Warnell database, providing the corporate sponsor with real-time, mathematically verified evidence of temperature cooling and spawning success.
3. **Positive Corporate ESG PR**: Instead of a defensive public relations posture, the data center operator sponsors a high-status project with **UGA Warnell** and **Trout Unlimited**. They receive a branded, scientifically validated conservation legacy: *"This Coldwater Spawning Sanctuary was fully funded by [Sponsor Name] in partnership with Blue Ridge Stream Restoration to protect native wild brook trout."*
4. **USACE-Approved Mitigation Credits**: Sponsoring our Nationwide Permit 27 geomorphic restorations generates official Savannah District stream mitigation credits that operators can use to satisfy regulatory requirements for their new campus construction, creating a highly efficient, dual-value investment.

Developed by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin under main.